CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

9.1 Electric Current

Syllabus Section: 9. Electricity **Paper Type:** Multiple Choice (Paper 1) **Total:** 20 Questions (20 Marks)

Question 1 Easy [1 mark]

When there is **no** current in a wire, which statement about the conduction electrons in that wire is correct?

A electrons in the wire are moving totally randomly within the wire

B equal numbers of electrons move at the same speed, but in opposite directions, along the wire

C no current is flowing therefore the electrons in the wire are stationary

D no current is flowing therefore the electrons in the wire are vibrating around a fixed point

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Even when there is no potential difference applied through a metal wire, the free electrons are still moving around rapidly
- Without the potential difference, the speed of the electrons is about 10^6 m s^{-1}
- However, since they are moving at random, there is no net flow of electrons in any particular direction and so there is no current
- When a voltage is applied, the electrons gain an additional velocity, so that there is a net flow along the wire

B is incorrect because when there is no current flow in a wire, free electrons in a metal move around with a range of speeds and travel in random directions, much like molecules of gas in the air

C is incorrect because the current is the flow of electrons, but when there is no potential difference, there is no net flow of electrons, this doesn't mean the electrons are stationary, it just means they are not all moving in a specific direction – this distinction is very important to learn

D is incorrect because the outer electrons in a metal are not confined to a fixed point, they are delocalised and are able to drift through the metal freely

A common misconception is that electrons orbit the individual metal atoms, but this is not true. In reality, the outermost electrons are orbiting among all the metal atoms all the time, in a sort of "sea of electrons". Another misconception is that electrons only jump between atoms when a voltage is applied along the wires, this is also not true. As we discussed in the question, the electrons in the "sea of electrons" are moving randomly inside the metal all the time. Applying the voltage causes the electrons to start drifting towards the positive terminal, which is what leads to a flow of electrons, better known as the current.

Question 2 Easy [1 mark]

When will 1 C of charge pass a point in an electrical circuit?

A when 1 A moves through a potential difference of 1 V

B when a power of 1 W is used for 1 s

C when the current is 5 mA for 200 s

D when the current is 10 A for 10 s

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Charge is defined by $Q = I \quad t$
- In option C, $I = 5 \times 10^{-3}$ A and $t = 200$ s
  - So, $Q = 5 \times 10^{-3} \times 200 = 1 \quad \text{C}$

A is incorrect because in order to calculate charge, we need current and time, current and potential difference alone aren't enough

B is incorrect because power and time can be used to calculate energy, not charge

D is incorrect because this gives $Q = 10 \times 10 = 100 \quad \text{C}$

Question 3 Easy [1 mark]

There is a current of 10 mA in a conductor for half an hour.

How much charge passes a point in the conductor in this time?

A 0.3 C

B 5 C

C 18 C

D 300 C

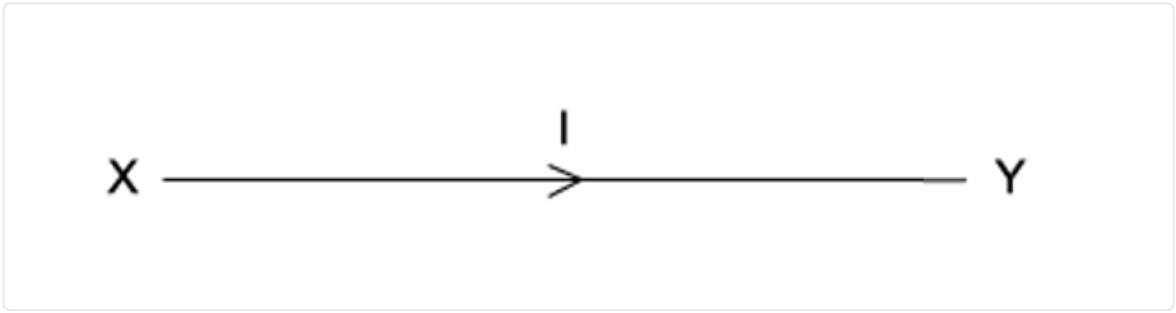
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Charge is defined by $Q = It$
  - Current *I* is equal to 10×10^{-3} A
  - The time *t* is equal to 30 minutes = 60 × 30 = 1800 s
- Charge is therefore equal to $Q = 10 \times 10^{-3} \times 1800 = 18 \quad \text{C}$

Question 4 Easy [1 mark]

The diagram shows the symbol for a wire carrying a current *I*



What does this current represent?

A the amount of charge flowing past a point in XY per second

B the number of electrons flowing past a point in XY per second

C the number of positive ions flowing past a point in XY per second

D the number of protons flowing past a point in XY per second

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Electric current is the rate of flow of charge
- It is defined as the amount of charge that passes through a point on a wire per unit time
- The only statement that matches this definition is option A

B, C & **D** are incorrect because electrons are the most common carriers of charge in electricity, but current is not specifically the flow of electrons, it can be the flow of any particle which carries a charge, whether it's electrons, protons or ions

Question 5 Easy [1 mark]

Which equation that links some of the following terms is correct?

potential difference (p.d.)	*V*
current	*I*
resistance	*R*
charge	*Q*
energy	*E*
power	*P*
time	*t*

A $P = \frac{Q^2 R}{t}$

B $ER^2 = V^2 t$

C $\frac{VI}{P} = t$

D $PQ = EI$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Power is equal to $P = \frac{E}{t}$
- Charge is equal to $Q = I \quad t$
- The units on the LHS
  - $PQ = WC = \text{J} \quad \text{s}^{-1}\text{A} \quad \text{s} = \text{J} \quad \text{A}$
- The units on the RHS
  - $EI = JA = \text{J} \quad \text{A}$

- The units on the LHS match the units on the RHS so option D is correct

A is incorrect because	LHS: $P = W = \text{J} \quad \text{s}^{-1}$ RHS: $\frac{Q^2R}{t} = \frac{I^2t^2}{t}\frac{V}{I} = VIt = E = \text{J}$	LHS ≠ RHS
B is incorrect because	LHS: $ER^2 = VIt\frac{V}{I}^2 = \frac{V^3t}{I}$ RHS: V^2t	LHS ≠ RHS
C is incorrect because	LHS: $\frac{VI}{P} = \frac{P}{P} = 1$ RHS: t	LHS ≠ RHS

Question 6 Easy [1 mark]

A copper wire of cross-sectional area 2.0 mm^2 carries a current of 10 A.

How many electrons pass through a given cross-section of the wire in one second?

A 1.0×10^1

B 5.0×10^6

C 6.3×10^{19}

D 3.1×10^{25}

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Current is defined as the amount of electric charge per unit of time
  - It is equal to $I = \frac{Q}{t}$
- First, we must find the charge Q over the period of 1 second:
  - $Q = It = 10 \times 1 = 10 \quad \text{C}$
- The number of electrons is calculated by dividing the total charge by the charge of one electron
- The charge of one electron e is 1.6×10^{-19} C
  - Number of electrons $= \frac{Q}{e} = \frac{10}{1.6 \times 10^{-19}} = 6.3 \times 10^{19}$

Question 7 Medium [1 mark]

Two copper wires of the same length but different diameters carry the same current.

Which statement about the flow of charged particles through the wires is correct?

A charged particles are provided by the power supply, therefore, the speed at which they travel depends only on the voltage of the supply

B the charged particles in both wires move with the same average speed because the current in both wires is the same

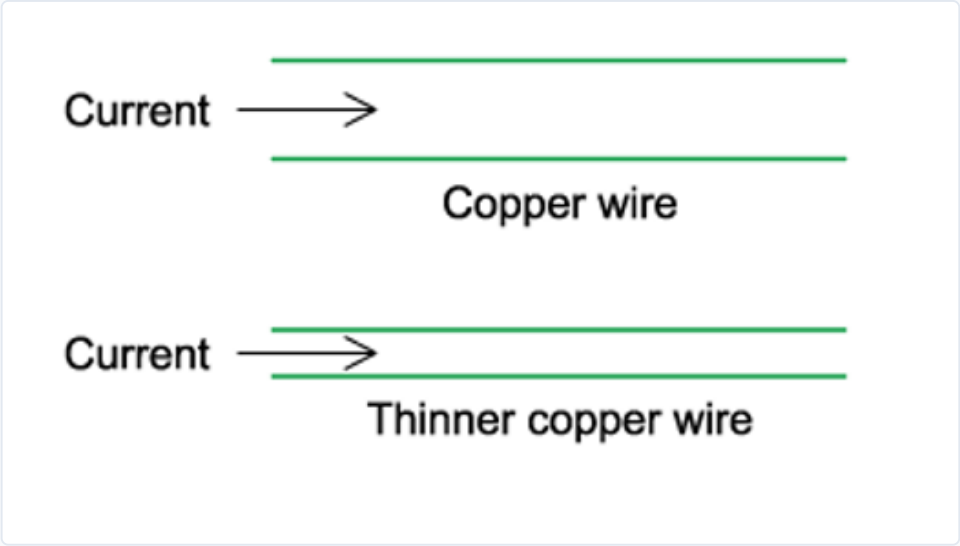
C the charged particles move faster through the wire with the larger diameter because there is a greater volume through which to flow

D the charged particles move faster through the wire with the smaller diameter because it has a larger potential difference applied to it

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The equation for current in a current-carrying conductor is given by
  - $I = nAve$
  - Rearranging this for *v* we find $v = \frac{1}{nAe}$
- Since the current *I*, number of electrons *n* and charge *e* are the same for both wires, the electrons are faster in the thinner (smaller *A*) wire
  - In the next topic we will look more at resistivity in a wire, for now let's just consider the equation $R = \frac{\rho L}{A}$
- Since length *L* and resistivity *ρ* are the same for both wires, the thicker (larger *A*) wire will have a smaller resistance, so, the thinner wire will have a larger resistance
- Current is equal to $I = \frac{V}{R}$ for each wire, and since the wires have the same current *I*, the value of $\frac{V}{R}$ will be the same in each wire
- This means the thinner wire will have a larger resistance *R* and a smaller potential difference *V* across it, while the thicker wire will have a smaller resistance *R* and a larger potential difference *V* across it
  - Therefore, the charged particles move faster through the thinner wire because it has a larger potential difference applied to it



A is incorrect because charged particles are not provided by the power supply, the charged particles (electrons) are already present in the copper wire

B is incorrect because the equation $\frac{1}{nAe}$ tells us the speed of the electrons depends on the thickness of the wire, so the speed will not be the same

C is incorrect because the equation $\frac{1}{nAe}$ tells us the electrons will be faster in the thinner wire (smaller *A*, larger *v*), not the thicker wire (larger *A*, smaller *v*)

Question 8 Medium [1 mark]

A milliammeter shows a reading of 20 mA.

How many electrons flow through the milliammeter in 10 seconds?

A 0.20

B 3.2×10^{20}

C 200

D 1.3×10^{18}

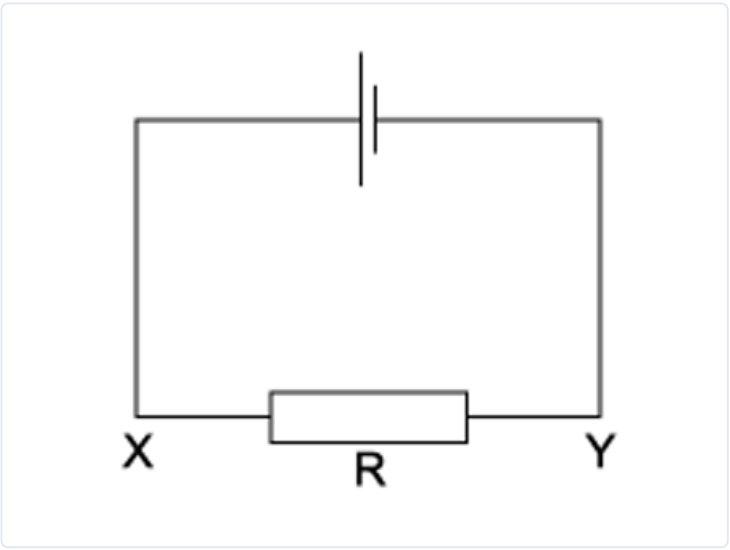
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Charge is defined by $Q = I \quad t$
- Current $I = 20 \times 10^{-3}\ \text{A}$
- Time $t = 10 \quad \text{s}$
- So, charge $Q = 20 \times 10^{-3} \times 10 = 0.2 \quad \text{C}$
- The total charge is equal to $Q = ne$
  - Where *n* is the number of electrons and *e* is the charge of one electron
- So, the number of electrons is equal to $n = \frac{Q}{e} = \frac{0.2}{1.6 \times 10^{-19}} = 1.25 \times 10^{18}$
  - Therefore, the number of electrons is 1.3×10^{18}

Question 9 Medium [1 mark]

The current in the circuit shown is 4.8 A.



What is the direction of flow and the rate of flow of electrons through the resistor *R* ?

	Direction of flow	Rate of flow
A	X to Y	$3.0 \times 10^{19}\ s^{-1}$
B	X to Y	$6.0 \times 10^{18}\ s^{-1}$
C	Y to X	$3.0 \times 10^{19}\ s^{-1}$
D	Y to X	$6.0 \times 10^{18}\ s^{-1}$

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Electrons flow from the negative terminal of the battery, since they are negatively-charged
- The negative terminal of the battery is represented by the shorter line at the symbol of the battery

- So, the direction of flow of the electrons is from Y to X

- In terms of current, charge is defined as $Q = I \quad t$
- In terms of number of electrons, charge is equal to $Q = ne$
  - Where *n* and *e* are the number and charge of electrons respectively
- So, $ne = I \quad t$ and the rate of flow of electrons is equal to $\frac{n}{t} = \frac{I}{e}$
  - Rate of flow of electrons $= \frac{n}{t} = \frac{I}{e} = \frac{4.8}{1.6 \times 10^{-19}} = 3.0 \times 10^{19} \quad \text{s}^{-1}$

Question 10 Medium [1 mark]

A wire with a resistance of 2.0 Ω has 8.2×10^{28} electrons pass through it at an average drift speed of 0.09 mm s^{-1}. The potential difference of the resistor is 12 V.

What is the cross-sectional area of the wire?

A 5.1 μm^2

B 0.14 mm^2

C 5.1 mm^2

D 0.14 cm^2

Mark Scheme & Explanation:

*The correct answer is **C** because*

- The known quantities are
  - Resistance, R = 2.0 Ω
  - Number of electrons, $n = 8.2 \times 10^{28}$
  - Potential difference, V = 12 V
  - Charge of an electron, $q = 1.6 \times 10^{-19}$ C (from the data sheet)
  - Average drift speed of electrons, v = 0.09 mm s^{-1} = 0.09×10^{-3} m s^{-1}
- The current through the resistor is
  - $V = IR$
  - $I = \frac{V}{R}$
  - $I = \frac{12}{2.0} = 6.0 \quad \text{A}$
- The cross-sectional area of the wire is
  - $I = Anvq$
  - $A = \frac{I}{nvq}$
  - $A = \frac{6.0}{8.2 \times 10^{28} \times 0.09 \times 10^{-3} \times 1.6 \times 10^{-19}} = 5.1 \quad \times 10^{-6}$ m^2 (2 s.f)
  - 10^{-6} m^2 = mm^2 therefore
  - A = 5.1 mm^2

- This is option **C**

Question 11 Medium [1 mark]

A 3.0 Ω resistor has 7.5×10^{18} electrons pass through it in 5.0 s.

What is the potential difference and charge of the resistor?

	Charge / C	Potential difference / V
A.	1.2	0.24
B.	1.2	0.72
C.	12	2.4
D.	12	7.2

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The known quantities are
  - Resistance, R = 3.0 Ω
  - Number of electrons, $n = 7.5 \times 10^{18}$
  - Time, t = 5.0 s
  - Charge of an electron, $e = 1.6 \times 10^{-19}$ C (from the data sheet)
- The charge through the resistor is
  - $Q = ne$
  - $Q = 7.5 \times 10^{18} \times 1.6 \times 10^{-19}$
  - $Q = 1.2$ C
  - This eliminates options **C** and **D**
- The current through the resistor is

- $I = \frac{Q}{t}$
- $I = \frac{1.2}{5.0} = 0.24$ A

- Therefore, the potential difference is
  - $V = IR$
  - V = 0.24 × 3.0 = 0.72 V
  - This means the answer is option **B**

Question 12 Medium [1 mark]

A beam of electrons, each carrying charge *e*, moves through a vacuum tube of uniform cross-sectional area *A* with uniform velocity *v*. The beam produces a current *I*.

What is the number density of electrons in the beam?

A $\frac{I}{Ae}$

B $\frac{I}{Ave}$

C $\frac{Iv}{Ae}$

D $\frac{I}{Av}$

Mark Scheme & Explanation:

The correct answer is B

- The electrons form a beam moving with uniform velocity *v* through the vacuum tube
- Current is related to charge flow by the equation

$$I = nAve$$

- Where:
  - *n* is the number density of electrons
  - *A* is the cross-sectional area of the vacuum tube
  - *v* is velocity
  - *e* is electron charge
- Rearranging for number density:

$$n = \frac{I}{Ave}$$

A is incorrect as it lacks the velocity term in the denominator

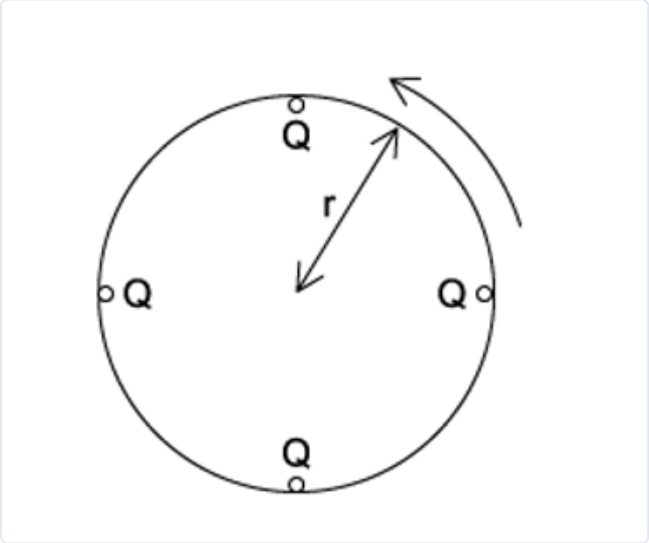
C is incorrect as velocity appears in the numerator rather than denominator

D is incorrect as it lacks the charge term in the denominator

Question 13 Hard [1 mark]

Four point charges, each of charge *Q*, are placed on the edge of an insulating disc of radius *r*.

The frequency of rotation of the disc is *f*.



What is the equivalent electric current at the edge of the disc ?

A $4Qf$

B $\frac{4Q}{f}$

C $8\pi rQ \quad f$

D $\frac{2Qf}{\pi r}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Charge is defined as $Q = I \quad t$
- The total charge on the disc is equal to $= Q + Q + Q + Q = 4Q$
- The frequency of rotation is equal to $f = \frac{1}{period \quad for \quad one \quad rotation} = \frac{1}{t}$
  - So, $t = \frac{1}{f}$
- Current is equal to $I = \frac{Q}{t}$
- Equivalent electric current $= \frac{total \quad charge \quad flow \quad per \quad rotation}{time \quad per \quad rotation} = \frac{4Q}{t} = 4Qf$

- Therefore, the equivalent electric current at the edge of the disc is $4Qf$.

Question 14 Hard [1 mark]

A parallel beam of protons, each with charge *e*, travel at uniform velocity *v*. This results in current *I* being formed.

What is the number of protons present in a unit length of the beam?

A $\frac{I}{e}$

B $\frac{I}{ev}$

C $\frac{I}{\frac{e}{v}}$

D $\frac{I}{v}$

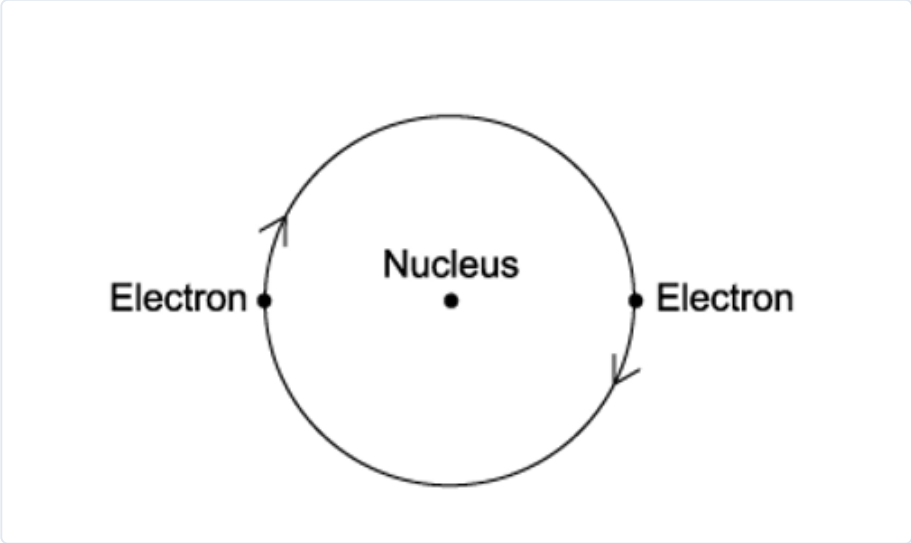
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The question mentions the protons travel as a parallel beam, this is a beam which does not diverge or converge – its width stays the same as the beam travels out from its source
- The equation for current in a current-carrying conductor is given by
  - $I = nAve$
  - Where *n* is the number density of protons and *A* is the cross-sectional area of the beam
- This can be written as number of protons per unit length
  - Since the unit of *n* is m^{-3}, the unit of *A* is m^2 and the unit of *N* is m^{-1}
- Rearranging this for *N* we find
  - $nA = N = \frac{1}{ev}$

Question 15 Hard [1 mark]

The diagram shows a model of an atom in which two electrons move round a circular orbit. The electrons complete 1.0×10^{15} revolutions around the nucleus every second.



What is the current caused by the motion of the electrons in the orbit?

A 1.6×10^{-34} A

B 3.2×10^{-34} A

C 1.6×10^{-4} A

D 3.2×10^{-4} A

Mark Scheme & Explanation:

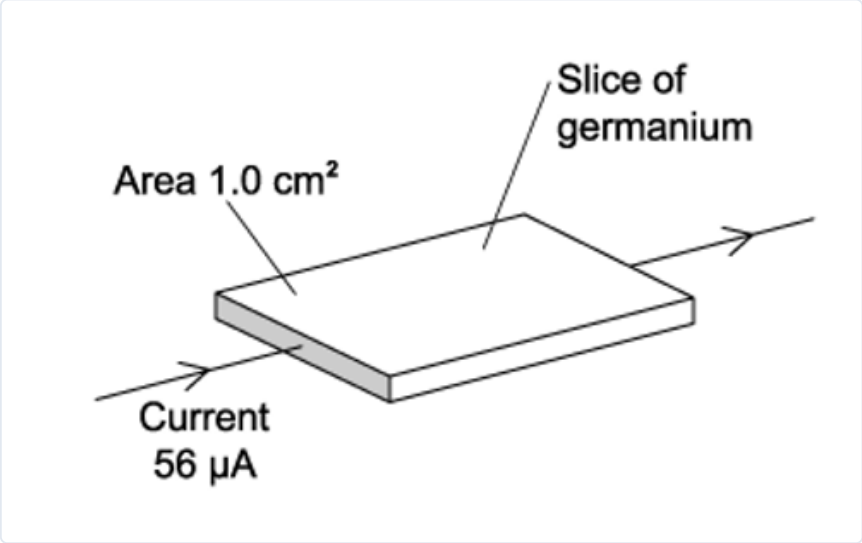
*The correct answer is **D** because:*

- Charge is defined by $Q = It$
- The charge on one electron is 1.6×10^{-19} C
- So, the total charge due to two electrons is $Q = 2 \times 1.6 \times 10^{-19}$
- The frequency of revolutions is $f = 1.0 \times 10^{15}$ Hz
- The time, or the period of one revolution, is given by $t = \frac{1}{f}$
  - So, $t = \frac{1}{1.0 \times 10^{15}} = 1.0 \times 10^{-15}$ s

- Therefore, the current is equal to $I = \frac{Q}{t} = \frac{2 \times 1.6 \times 10^{-19}}{1.0 \times 10^{-15}} = 3.2 \times 10^{-4}$ A

Question 16 Hard [1 mark]

A slice of germanium of cross-sectional area 1.0 cm^2 carries a current of 56 μA. The number density of charge carriers in the germanium is 2.0 × 10^{13} cm^{-3}.



Each charge carrier has a charge equal to the charge on an electron.

What is the average drift velocity of the charge carriers in the germanium?

A 0.18 m s^{-1}

B 18 m s^{-1}

C 180 m s^{-1}

D 1800 m s^{-1}

Mark Scheme & Explanation:

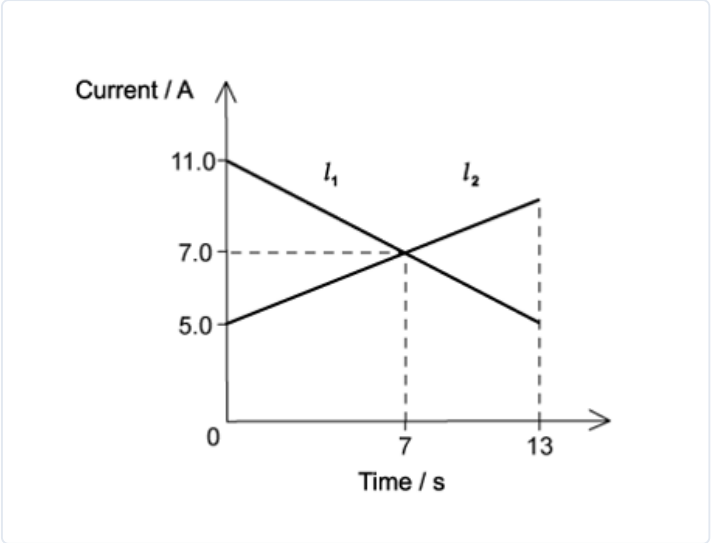
*The correct answer is **A** because:*

- Current in a current-carrying conductor is equal to
  - $I = nqAv$
- The drift velocity is found by rearranging for v
  - $v = \frac{I}{nqA}$
- First, we need to convert the quantities given into SI units.

- Cross-sectional area $A = 1.0 \quad \text{cm}^2 = 1.0 \times 10^{-2\,2} \quad \text{m}^2 = 1.0 \times 10^{-4} \quad \text{m}^2$
- Number density $n = 2.0 \times 10^{13}\,\text{cm}^{-3}$
- This means, in every 1 cm^3, there are 2.0 × 10^{13} charge carriers
    - $1 \quad cm^3 = 1 \times 10^{-2\,3} \quad \text{m}^3 = 1 \times 10^{-6} \quad \text{m}^3$
- So, in every 1 m^3, there are $\frac{2.0 \times 10^{13}}{1 \times 10^{-6}} = 2.0 \times 10^{19} \quad \text{m}^{-3}$ charge carriers
- So, number density $n = 2.0 \times 10^{19} \quad \text{m}^{-3}$
- The charge of one electron is $q = 1.6 \times 10^{-19} \quad \text{C}$
    - Drift velocity $v = \frac{I}{nqA}$
    - Drift velocity $v = \frac{56 \times 10^{-6}}{2.0 \times 10^{19}\,1.0 \times 10^{-4}\,1.6 \times 10^{-19}} = 0.175 \quad \text{m} \quad \text{s}^{-1}$
- Therefore, the average drift velocity of the charge carriers is 0.18 m s^{-1}

Question 17 Hard [1 mark]

I_1 and I_2 are currents flowing through two different circuits, as shown in the diagram below.



Calculate the difference in the total charge that flows through the two circuits in the first 7 seconds.

A 0 C

B 21 C

C 42 C

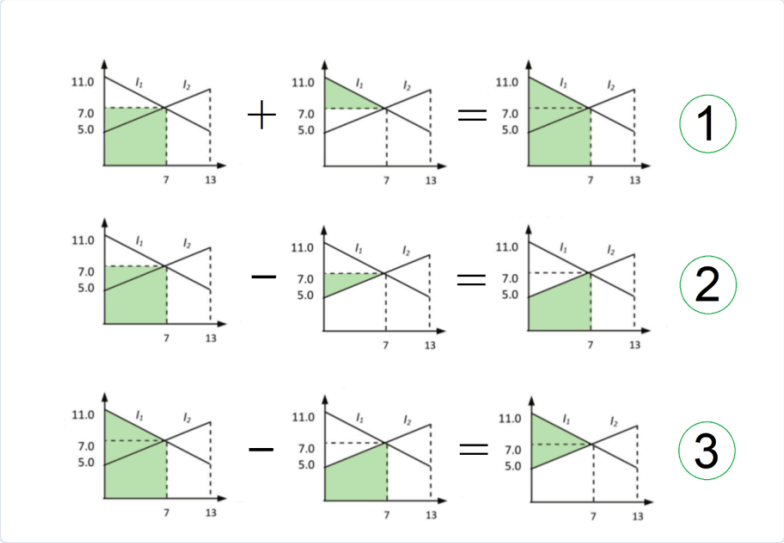
D 49 C

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Charge is defined by $Q = I \quad t$
- Charge can be calculated from the area under a current-time graph
- We need to find the total charge that flows through the two circuits up to 7 seconds on the graph
  - Difference in total charge that flows through the two circuits = Difference in area under graphs for the first 7 seconds
- To work this out we will need three steps (as illustrated by the diagram)
  - Step 1: Charge through $I_1 = 7 \times 7 + \frac{1}{2}11 - 77 = 63 \quad \text{C}$
  - Step 2: Charge through $I_2 = 7 \times 7 - \frac{1}{2}7 - 57 = 42 \quad \text{C}$

- Step 3: Difference in charges $= 63 - 42 = 21 \quad C$



Exam Tip: a quicker way to do this problem would be to just find the area shown in green in the last graph in step 3, can you see how this could be done?

Question 18 Hard [1 mark]

The Atlantic torpedo is a large electric fish capable of generating a voltage of 220 V between its tail and its head. This drives a pulse of current of 15 A lasting for a time of 2.0 ms. The fish produces 200 pulses per second.

What is the average power output of the fish?

A 33 W

B 1.3 kW

C 3.3 kW

D 6.6 kW

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Electric power is equal to $P = VI$
- Let's take another definition of power $P = \frac{E}{t}$
- We can combine these two equations to give $P = VI = \frac{E}{t}$
  - So, energy is equal to $E = VIt$
- A pulse of current of 15 A lasts for 2.0 ms
- Energy of each pulse $= 220 \times 15 \times 2.0 \times 10^{-3} = 6.6 \text{ J}$
- Average power $= \frac{total\ \ energy}{time}$
- Average power $= no.of\ \ pulses \times \frac{energy\ \ of\ \ 1\ \ pulse}{time}$
- Average power $= \frac{no.of\ \ pulses}{time} \times energy\ \ of\ \ 1\ \ pulse$
- The fish produces 200 pulses per second
  - We can write this as $\frac{no.of\ \ pulses}{time} = 200\ \ s^{-1}$
- Average power $= \frac{no.of\ \ pulses}{time} \times energy\ \ of\ \ 1\ \ pulse$
  - Average power $= 200 \times 6.6 = 1320 \text{ W}$
- Therefore, the average power output of the fish is 1.3 kW

Question 19 Hard [1 mark]

A high potential is applied between the electrodes of an ionised gas. The gas carries a current of 8.16 mA and the number of electrons passing any point in the gas per unit time is 2.58×10^{16} s^{-1}

If the charge on each positive particle is 3.2×10^{-19} C, what is the number of positively charged particles passing any point in the gas per unit time?

A 1.26×10^{16} s^{-1}

B 2.58×10^{16} s^{-1}

C 3.84×10^{16} s^{-1}

D 10.3×10^{16} s^{-1}

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- When a potential difference is applied to the ionised gas, the positive charges move towards the negative terminal and the negative charges (electrons) move towards the positive terminal
- The total current is given by the sum of these currents
  - $I = I^{+} + I^{-}$
- Current is equal to $I = \frac{total\ \ charge}{time} = \frac{Q}{t}$
- The total charge Q is equal to the sum of charges each with charge q
  - This can be expressed as $Q = Nq$
- The current of the positive charges is given by $I^{+} = \frac{Nq^{+}}{t}$
  - Where q = 3.2×10^{-19} C
- The current of the negative charges is given by $I^{-} = \frac{Nq^{-}}{t}$
  - Where q = 1.6×10^{-19} C and $\frac{N}{t}$ = 2.58×10^{16} s^{-1}
- We can combine these equations to give
  - $I = \frac{Nq^{+}}{t} + \frac{Nq^{-}}{t}$
  - $8.16 \times 10^{-3} = \frac{N}{t} 3.2 \times 10^{-19} + 2.58 \times 10^{16} 1.6 \times 10^{-19}$
  - $\frac{N}{t} = \frac{8.16 \times 10^{-3} - 4.13 \times 10^{-3}}{3.2 \times 10^{-19}} = 1.26 \times 10^{16}$ s^{-1}

- Therefore, the number of positive charges passing a point is 1.26×10^{16} s^{-1}

Question 20 Hard [1 mark]

An electron beam of current 50 μA flows in a cathode-ray oscilloscope. The time-base causes the beam to sweep horizontally across the screen at 1.0×10^{4} cm s^{-1}

What is the number of electrons arriving at the screen in one centimetre length of the horizontal trace?

A 3.1×10^{10}

B 3.1×10^{13}

C 1.3×10^{13}

D 8.0×10^{14}

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Charge is defined by $Q = It$
- The total charge is equal to $Q = ne$
  - Where *n* is the number of electrons and *e* is the charge of one electron
- If we equate these two equations, we obtain $ne = It$
  - So, the number of electrons is given by $n = \frac{It}{e}$
- First, we must find the time taken to produce the 1 cm trace
  - $t = \frac{1}{1.0 \times 10^{4}} = 1.0 \times 10^{-4}$ s
- We know the current is $I = 50 \ \mu A = 50 \times 10^{-6}$ A
  - And the charge on an electron is $e = 1.6 \times 10^{-19}$ C
- So, the number of electrons arriving at the trace is
  - $n = \frac{It}{e} = \frac{50 \times 10^{-6} \, 1.0 \times 10^{-4}}{1.6 \times 10^{-19}} = 3.125 \times 10^{10}$
- Therefore, the number of electrons is 3.1×10^{10}